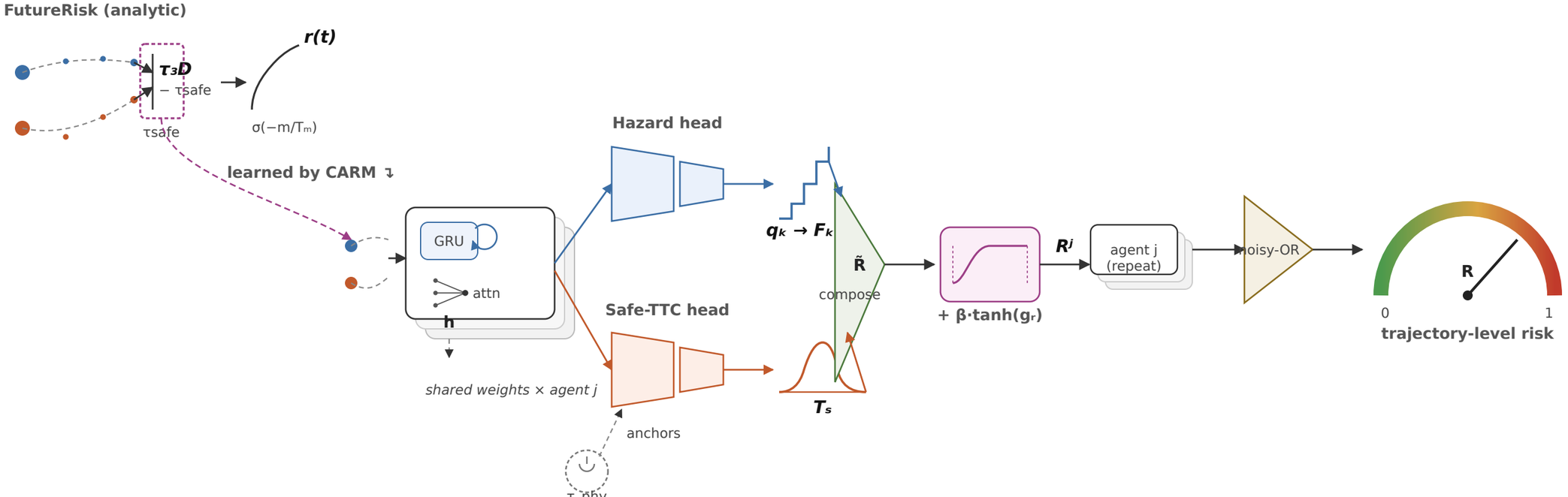

● ego ● agent j - - observed track hazard / event-time safe-TTC (physics-anchored) bounded residual